Canadian Immigration Statistics — Data Quality Report

Internal-consistency checks on processed IRCC/IRB tables

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1 Purpose and methodology

This report checks the internal arithmetic consistency of the 89 tidy tables produced by this project's ingest pipeline from IRCC's and IRB's open data. Two independent families of checks are run on every processed table:

Column-wise, within a row — for a single row category (e.g. Ontario, Female), do its own figures add up across time?

- Do a quarter's three monthly values sum to its published quarter total?

- Do a year's monthly values sum to its published annual total (where a table publishes both)?
- Do a year's quarterly totals sum to its published annual total (where a table publishes both)?

Row-wise, within a period column — for a single point in time (e.g. January 2015), do a table's nested row categories add up correctly?

- Do a group's detail rows (e.g. the provinces within a group) sum to the group's published subtotal row?
- Do a table's top-level subtotal rows sum to its published grand total?

Row-wise checks only apply to tables with a nested row hierarchy (IRCC's "pivot" Excel tables); flat, already-tidy tables have nothing to check here since they have no subtotal rows.

1.1 Why exact equality isn't the bar

IRCC and IRB round every published figure to the nearest multiple of 5, independently at every level of a table, and suppress ("--") any value between 0 and 5 to protect privacy. That means even a perfectly correct table's components essentially never sum to *exactly* their published total — only to within rounding and suppression noise. A check that required exact equality would flag nearly everything, which would be useless.

Instead, each comparison is scored against a conservative tolerance band:

$$\text{tolerance} = 2.5 \times (n_{\text{terms}} + 1) + 5 \times n_{\text{suppressed}}$$

- $2.5 \times (n_{\text{terms}} + 1)$ allows for each summed component's own rounding error (up to ± 2.5) plus the total/subtotal cell's own rounding error.
- $5 \times n_{\text{suppressed}}$ allows for each suppressed ("--") component, which could be as large as 5 but is treated as 0 in the observed sum.

A comparison is marked:

- **pass** — the difference between the published total and the sum of its parts is within tolerance.
- **anomaly** — the difference exceeds tolerance, flagged for review.
- **not checkable** — the total/subtotal cell itself is suppressed, so there is nothing to compare against.
- **not applicable** — the check doesn't apply to this table (see "stock" tables below).

Checks by family and outcome

family	anomaly	not_applicable	not_checkable	pass
Period (column-wise)	5,578	31,911	335,665	1,326,663
Row (row-wise)	4,937	0	118,359	615,454

Checks by type and outcome

check_type	anomaly	pass	not_applicable	not_checkable
grand_total	560	9648	0	0
month_to_quarter	5578	1326627	31899	335665
month_to_year	0	24	0	0
quarter_to_year	0	12	12	0
row_subtotal_level_1	1405	122479	0	12443
row_subtotal_level_2	101	244211	0	61867
row_subtotal_level_3	2868	234965	0	43159
row_subtotal_level_4	3	4151	0	890

2 Overall results

Of 1,952,632 checks where the published total was not itself suppressed, 10,515 (0.539%) fell outside the rounding/suppression tolerance and are flagged as anomalies below.

3 Anomalies by table

4 Anomaly detail

Several source tables cover the same underlying data under different resource names (a known gap in the manifest dedup - see the project README), so the same anomaly often appears identically in more than one processed table. The list below is deduplicated by its numeric signature (check type, period, published total, sum of parts) so each *distinct* anomaly is shown once, with a count of how many tables exhibit it and one example table:

Tables with the most flagged anomalies (top 25)

dataset	table	family	Anomalies
temporary_workers	canada international mobility program…	Row (row-wise)	3311
temporary_workers	canada international mobility program…	Period (column-wise)	1819
study_permit_holders	canada post secondary by p t of desig…	Period (column-wise)	1118
study_permit_holders	canada study permit holders by provin…	Period (column-wise)	1111
study_permit_holders	canada study permit holders by provin…	Row (row-wise)	1065
temporary_workers	canada work permit holders for humani…	Period (column-wise)	725
temporary_workers	canada temporary foreign worker progr…	Period (column-wise)	562
study_permit_holders	canada study permit holders by countr…	Period (column-wise)	243
temporary_workers	canada work permit holders for humani…	Row (row-wise)	211
study_permit_holders	canada post secondary by p t of desig…	Row (row-wise)	205
temporary_workers	canada temporary foreign worker progr…	Row (row-wise)	82
study_permit_holders	canada study permit holders with a va…	Row (row-wise)	63

5 Two investigated sources of anomalies

Both patterns below were found by tracing the largest flagged anomalies back to their source spreadsheets. Both are validator-methodology mismatches rather than errors in IRCC's or IRB's published figures.

5.1 1. "Valid permit" (stock) tables aren't additive across months - fixed, not just flagged

Most IRCC tables measure a *flow*: permits that became effective in a given month, which are naturally additive (Jan + Feb + Mar = Q1). But 5 tables - "Study permit holders **with a valid permit** by province/territory…" and its siblings - measure a *stock*: how many permit holders had a currently-valid permit as of that month. Stock counts are not additive across time (the same person holding a valid permit in January, February, and March would be triple-counted), so the "Q# Total" column in these tables is not a sum of its three months - checking one against the other in the same way naturally produces a large, consistent mismatch. Spot-checking a specific row (Alberta / Albania, 2018 Q2) shows the mechanism directly: Apr = 15, May = 15, Jun = 15, **Q2 Total = 15** - the quarter figure equals each individual month, not their sum.

This is not a data quality problem - it means the `month_to_quarter` / `quarter_to_year` checks simply don't apply to stock-type tables. Rather than guess from table titles (not every stock table's name says "valid permit" - two of the five affected here are older legacy exports that don't), `R/validate_utils.R::reclassify_stock_tables()` detects the pattern directly from the numbers: any table whose `month_to_quarter` sum-of-parts runs consistently ~3x its published total gets those checks relabelled **not applicable** rather than left as false-positive

Largest distinct anomalies (top 25 by signature)

Check	Period	Published total	Sum of parts	Difference	# tables	Example table
grand_total	2024 2024 Total	180,085	360,245	-180,160	1	canada work permit holders for humani…
grand_total	2025 2025 Total	158,265	316,750	-158,485	1	canada work permit holders for humani…
grand_total	2023 2023 Total	130,765	261,560	-130,795	1	canada work permit holders for humani…
grand_total	2022 2022 Total	75,125	150,320	-75,195	1	canada work permit holders for humani…
grand_total	2026 2026 Total	67,255	134,510	-67,255	1	canada work permit holders for humani…
grand_total	2026 Q1 Q1 Total	63,505	127,030	-63,525	1	canada work permit holders for humani…
grand_total	2019 2019 Total	62,735	125,500	-62,765	1	canada work permit holders for humani…
grand_total	2021 2021 Total	51,500	103,020	-51,520	1	canada work permit holders for humani…
grand_total	2018 2018 Total	50,880	101,835	-50,955	1	canada work permit holders for humani…
grand_total	2025 Q4 Q4 Total	49,485	99,005	-49,520	1	canada work permit holders for humani…
grand_total	2024 Q2 Q2 Total	49,470	98,935	-49,465	1	canada work permit holders for humani…
grand_total	2020 2020 Total	47,890	95,825	-47,935	1	canada work permit holders for humani…
grand_total	2024 Q3 Q3 Total	46,975	93,945	-46,970	1	canada work permit holders for humani…
grand_total	2024 Q1 Q1 Total	43,650	87,305	-43,655	1	canada work permit holders for humani…
grand_total	2025 Q1 Q1 Total	43,465	86,945	-43,480	1	canada work permit holders for humani…
grand_total	2024 Q4 Q4 Total	43,105	86,230	-43,125	1	canada work permit holders for humani…
grand_total	2017 2017 Total	38,140	76,330	-38,190	1	canada work permit holders for humani…
grand_total	2025 Q2 Q2 Total	36,695	73,405	-36,710	1	canada work permit holders for humani…
grand_total	2025 Q3 Q3 Total	36,000	72,020	-36,020	1	canada work permit holders for humani…
grand_total	2023 Q4 Q4 Total	35,620	71,245	-35,625	1	canada work permit holders for humani…
grand_total	2023 Q3 Q3 Total	34,625	69,235	-34,610	1	canada work permit holders for humani…
grand_total	2023 Q2 Q2 Total	32,705	65,405	-32,700	1	canada work permit holders for humani…
grand_total	2023 Q1 Q1 Total	29,030	58,055	-29,025	1	canada work permit holders for humani…
grand_total	2026 Q1 Feb	27,065	54,140	-27,075	1	canada work permit holders for humani…
grand_total	2022 Q4 Q4 Total	24,005	48,000	-23,995	1	canada work permit holders for humani…

anomalies. 31,911 checks across 5 tables were reclassified this way - without this fix, they would have made up 75.2% of everything flagged in this report.

5.2 2. Appended second breakdowns (row-wise)

Separately, some IRCC Excel sheets append a **second, independently-grouped breakdown** below the main table - sharing the same grand total - rather than publishing it as a separate table. For example, "Work permit holders for Humanitarian and Compassionate purposes by gender, age…" lists a full gender → age breakdown first, then appends a second, flat age-only breakdown underneath, both summing to the same overall total. The current parser does not detect this second-table boundary, so it treats the appended rows as more members of the same row hierarchy - which mechanically causes a subtotal/grand-total check to see roughly double the true count.

194 of the 10208 grand-total checks (across 1 table(s)) show almost exactly a 2x sum-to-total ratio, the signature of this pattern. The remaining flagged row-wise anomalies - particularly the elevated rate at `row_subtotal_level_1` and `row_subtotal_level_3` shown above - warrant a manual look against their source spreadsheets; some may be further instances of this same appended-table pattern (in tables with deeper nesting, an appended second breakdown surfaces at whichever row level it was inserted at, not necessarily level 1), and some may be genuine artifacts worth tracing further. This report flags them; it does not attempt to auto-correct them.

6 Coverage notes

- `not_checkable` results (total/subtotal itself suppressed) and `not_applicable` results (stock-type tables, see above) are both excluded from the anomaly rate above since there is nothing to verify in either case - neither is evidence of a problem.
- Row-wise checks only run against tables with a nested row hierarchy (IRCC's pivot-format Excel tables). Flat/already-tidy tables (most CSVs, and both IRB tables) have no subtotal rows to check and so contribute no row-wise results.
- Column-wise year-level checks (`month_to_year`, `quarter_to_year`) only run where a table actually publishes an annual total alongside its monthly/quarterly breakdown, which most IRCC tables don't (the current, still-in-progress year is the main exception).
- Row-wise anomaly rates vary noticeably by nesting level (see the check-type table above) - levels 2 and 4 are consistently clean while levels 1 and 3 run higher, concentrated in a handful of tables rather than spread evenly. That unevenness is itself a useful signal for where to look first.